

Precision Placement in Scaled Dot-Product Attention

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Abstract. We study scaled dot-product attention as a precision-placement problem. Rather than asking whether one global low-precision dtype is acceptable, we ask which individual stages of the forward pass can safely run in low precision: storage of the query, key, and value tensors, the logit product QK^T , logit storage, the rowwise softmax, and the final value product. The present draft still analyzes an analytically transparent sketch-based baseline, but its main purpose is to motivate a broader decomposition of attention error into logit perturbation, softmax perturbation, and final output perturbation. The accompanying code now includes exact-attention precision sweeps with policy-level controls and diagnostics for relative logit error, rowwise softmax error, and output error.

Keywords: mixed-precision arithmetic, attention, transformers, numerical stability, floating-point arithmetic

1 Introduction

Scaled dot-product attention [1] is the core computational primitive of transformer architectures. As models grow, the memory and compute cost of the attention forward pass motivates the use of reduced-precision arithmetic. The natural question is a *precision-placement* question: which stages of the forward pass can safely run in low precision, and which still require higher-precision accumulation or reduction?

The relevant stages are: storage of the query, key, and value tensors; the logit product QK^T ; logit storage; the rowwise softmax; and the final value product. A global dtype comparison—the usual approach—conflates these stages and cannot identify which *locations* are numerically sensitive. This report instead decomposes the attention pipeline and asks, for each stage separately, whether low precision there produces acceptable error.

The analytical core is Proposition 3, a local perturbation bound for exact attention that separates the effect of logit perturbations (weighted by $\|V\|_2$) from value perturbations (weighted by $\sqrt{n}$ from the row-stochastic spectral norm). This decomposition directly predicts which precision locations are sensitive: logit errors propagate through softmax with unit Lipschitz constant, while value errors propagate with a $\sqrt{n}$ factor that is moderate for the small sequence lengths tested here, but potentially important for long contexts. The sharpest failure mode is therefore low-precision softmax: when Σ itself is computed in low precision, the Lipschitz guarantee from Proposition 2 no longer

holds, and the effective perturbation of the probability matrix can be much larger than $\|\Delta L\|_{\text{HS}}$.

Sketching-based attention approximations [2] [3] [4] [5] [6] are retained as a methodological baseline: they provide a controlled stress test in which the error decomposition can be verified under a known approximation floor. But they are not the main contribution.

Although the experiments below focus mainly on floating-point precision policies, the same placement question also covers quantization. In quantized inference, the choice is not only which dtype to use, but also which tensors to quantize, how to choose scales and zero-points, whether calibration is dynamic or static, and which reductions must remain in higher precision. We therefore treat quantization as an extension of the same local-perturbation framework.

2 Operator-Theoretic Setup

Let $\mathcal{H}_d := \mathbb{R}^d$ and $\mathcal{H}_n := \mathbb{R}^n$, both finite-dimensional Hilbert spaces endowed with their standard Euclidean inner products. We view matrices in $\mathbb{R}^{n \times d}$ as Hilbert–Schmidt operators

$$Q, K, V : \mathcal{H}_d \rightarrow \mathcal{H}_n.$$

Their Hilbert–Schmidt norm is denoted by $\|\cdot\|_{\text{HS}}$, while $\|\cdot\|_2$ denotes the spectral norm on linear operators.

The exact scaled logit matrix, viewed as an operator on $\mathcal{H}_n$, is

$$\mathcal{L}(Q, K) := d^{-1/2} Q K^* \in \mathcal{L}(\mathcal{H}_n),$$

and the rowwise softmax map is denoted by

$$\Sigma : \mathbb{R}^{n \times n} \rightarrow \mathbb{R}^{n \times n}.$$

Accordingly, the attention output is the nonlinear operator

$$\mathcal{A}(Q, K, V) := \Sigma(\mathcal{L}(Q, K))V.$$

3 Analytical Core

3.1 Mixed Precision versus Quantization

In this report we distinguish mixed precision from quantization. Mixed precision changes the floating-point format used at different stages of a computation, for example storing Q, K, V in `float16` or `bfloat16` while accumulating QK^T and computing the softmax in `float32`. Quantization is more explicit: a real-valued tensor X is mapped to a discrete low-bit representation and then, for computation, usually dequantized back to a floating-point format.

A common affine quantization model is

$$x \approx S(x_q - Z),$$

where x_q is the quantized integer value, $S > 0$ is a scale, and Z is a zero-point chosen so that real zero can be represented exactly. Equivalently,

$$x_q = \text{clip}(\text{round}(x/S + Z)).$$

Symmetric quantization corresponds to the special case $Z = 0$ after choosing a symmetric representable range. The scale and zero-point may be chosen per tensor or per channel, and activation ranges may be determined dynamically, by static calibration, or during quantization-aware training.

For attention, quantization is another precision-placement mechanism. One may quantize Q, K, V , the logits $L = QK^T/\sqrt{d}$, the attention probabilities $\Sigma(L)$, or the final output. The perturbation framework below treats quantization by writing

$$\hat{X} = X + \Delta_{\text{quant}}X$$

for each quantized tensor, where $\Delta_{\text{quant}}X = D(Q(X)) - X$. Thus quantization errors enter the same local error bound as floating-point storage and rounding errors, but their structure is different: they depend on calibration range, clipping, scale granularity, and dequantization policy.

Mechanism	Typical form	Main error source
Mixed precision	fp16/bf16/fp8 storage, fp32 accumulation	floating-point rounding and overflow
Affine quantization	$x \approx S(x_q - Z)$	rounding, clipping, calibration error
Symmetric quantization	$x \approx Sx_q$	rounding and clipping under symmetric range
Dynamic activation quantization	runtime range estimation	range-estimation overhead and variance
Static activation quantization	calibrated range from representative data	calibration mismatch
Quantization-aware training	fake quantization during training	training adapts to quantization noise

Table 1 / Mixed precision and quantization as precision-placement mechanisms.

The analysis begins with a stability property of the rowwise softmax map, which is used in both the exact-attention and sketching bounds.

Proposition 2 (Rowwise softmax is Hilbert–Schmidt Lipschitz). Let $L, M \in \mathbb{R}^{n \times n}$. Then

$$\|\Sigma(L) - \Sigma(M)\|_{\text{HS}} \leq \|L - M\|_{\text{HS}}.$$

Consequently, for every $V \in \mathbb{R}^{n \times d}$,

$$\|\Sigma(L)V - \Sigma(M)V\|_{\text{HS}} \leq \|L - M\|_{\text{HS}} \|V\|_2.$$

Proof. Let $\sigma : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be the softmax map on vectors with Jacobian $D\sigma(x) = \text{Diag}(p) - pp^T$ where $p = \sigma(x)$. For any $z \in \mathbb{R}^n$, $z^T D\sigma(x) z = \sum_i p_i z_i^2 - (\sum_i p_i z_i)^2 \leq \|z\|_2^2$, so $\|D\sigma(x)\|_2 \leq 1$ for all x . The mean value theorem gives $\|\sigma(x) - \sigma(y)\|_2 \leq \|x - y\|_2$; applying this row by row and squaring yields the HS bound. The second assertion follows from $\|XV\|_{\text{HS}} \leq \|X\|_{\text{HS}} \|V\|_2$. $\square$

3.2 Exact-Attention Perturbation Bound

The next proposition is the main analytical result. It decomposes the attention error into a logit-driven term and a value-driven term, directly predicting the relative sensitivity of different precision placements.

Proposition 3 (Exact-attention local perturbation bound). Let

$$A = \Sigma(L)V, \quad \hat{A} = \Sigma(\hat{L})\hat{V},$$

where $L, \hat{L} \in \mathbb{R}^{n \times n}$ and $V, \hat{V} \in \mathbb{R}^{n \times d}$. Writing

$$\Delta L := \hat{L} - L, \quad \Delta V := \hat{V} - V,$$

one has

$$\|A - \hat{A}\|_{\text{HS}} \leq \|\Delta L\|_{\text{HS}} \|V\|_2 + \|\Sigma(\hat{L})\|_2 \|\Delta V\|_{\text{HS}}.$$

In particular, since $\Sigma(\hat{L})$ is row-stochastic and therefore $\|\Sigma(\hat{L})\|_2 \leq \sqrt{n}$,

$$\|A - \hat{A}\|_{\text{HS}} \leq \|\Delta L\|_{\text{HS}} \|V\|_2 + \sqrt{n} \|\Delta V\|_{\text{HS}}.$$

Proof. Add and subtract $\Sigma(\hat{L})V$ and apply the triangle inequality:

$$\|A - \hat{A}\|_{\text{HS}} \leq \|\Sigma(L)V - \Sigma(\hat{L})V\|_{\text{HS}} + \|\Sigma(\hat{L})(V - \hat{V})\|_{\text{HS}}.$$

The first term is bounded by Proposition 2:

$$\|\Sigma(L)V - \Sigma(\hat{L})V\|_{\text{HS}} \leq \|L - \hat{L}\|_{\text{HS}} \|V\|_2.$$

For the second,

$$\|\Sigma(\hat{L})(V - \hat{V})\|_{\text{HS}} \leq \|\Sigma(\hat{L})\|_2 \|\Delta V\|_{\text{HS}}.$$

Combining these gives the stated bound. $\square$

Remark 4 (Interpretation for precision placement). Proposition 3 directly predicts the relative danger of different precision choices:

- (1) **Logit perturbations** ($\Delta L \neq 0, \Delta V = 0$): The error scales as $\|\Delta L\|_{\text{HS}} \|V\|_2$. Exact softmax is Lipschitz and therefore does not amplify logit perturbations in the HS/L2 sense. However, the diagnostic ratio ρ_{softmax} , which compares a rowwise ℓ_∞ logit error to an L1 probability error, can become large when the logit perturbation denominator is extremely small or when the softmax computation itself is rounded in low precision—a regime the experiments expose.
- (2) **Value perturbations** ($\Delta L = 0, \Delta V \neq 0$): The error scales as $\sqrt{n} \|\Delta V\|_{\text{HS}}$. This $\sqrt{n}$ factor is moderate for the small sequence lengths tested here, but potentially important for long contexts.
- (3) **Low-precision softmax**: When Σ itself is computed in low precision, the Lipschitz bound from Proposition 2 no longer applies, and the effective perturbation of the probability matrix can be much larger than $\|\Delta L\|_{\text{HS}}$. Low-precision softmax thus introduces additional probability-matrix error beyond the perturbation predicted by $\triangleright$

exact-softmax Lipschitz stability. The experiments confirm this: ρ_{softmax} spikes to $\sim 10^3$ on transformer activations when softmax is run in bf16/fp16, while remaining below 1 on Gaussian data and only order-unity for safe transformer policies.

3.3 Supporting Background: Sketching as a Controlled Baseline

The following propositions analyze a sketched surrogate $\tilde{\mathcal{A}}_S(Q, K, V) = \Sigma(d^{-1/2}QSS^*K^*)V$ obtained by replacing the full logit product with a low-dimensional projection through $S \in \mathbb{R}^{d \times s}$ with $s < d$. These results are retained as a methodological baseline: the sketching error provides a known approximation floor against which the finite-precision error can be isolated and studied.

Proposition 5 (Deterministic sketching bound). Let $U \subseteq \mathcal{H}_d$ be the joint row space of Q and K , equivalently the subspace spanned by the columns of $[Q^T \ K^T]$, and let $r := \dim U$. Assume that S is an ϵ -subspace embedding for U , meaning

$$|\langle S^*u, S^*v \rangle - \langle u, v \rangle| \leq \epsilon \|u\|_2 \|v\|_2 \quad \text{for all } u, v \in U.$$

Equivalently, if $U_0 \in \mathbb{R}^{d \times r}$ is any orthonormal basis matrix for U , then $\|U_0^T SS^*U_0 - I_r\|_2 \leq \epsilon$. Then

$$\|\mathcal{L}(Q, K) - \tilde{\mathcal{L}}_S(Q, K)\|_{\text{HS}} \leq \frac{\epsilon}{\sqrt{d}} \|Q\|_{\text{HS}} \|K\|_2.$$

Consequently,

$$\|\mathcal{A}(Q, K, V) - \tilde{\mathcal{A}}_S(Q, K, V)\|_{\text{HS}} \leq \frac{\epsilon}{\sqrt{d}} \|Q\|_{\text{HS}} \|K\|_2 \|V\|_2.$$

Proof. Let $U_0 \in \mathbb{R}^{d \times r}$ have orthonormal columns spanning the joint row space of Q and K . Write $Q = Q_U U_0^T$ and $K = K_U U_0^T$, where $Q_U = Q U_0 \in \mathbb{R}^{n \times r}$ and $K_U = K U_0 \in \mathbb{R}^{n \times r}$. Then

$$QK^* - QSS^*K^* = Q_U(I_r - U_0^T SS^*U_0)K_U^T.$$

The subspace-embedding condition is exactly $\|U_0^T SS^*U_0 - I_r\|_2 \leq \epsilon$, so

$$\|QK^* - QSS^*K^*\|_{\text{HS}} \leq \|Q_U\|_{\text{HS}} \|I_r - U_0^T SS^*U_0\|_2 \|K_U^T\|_2 \leq \epsilon \|Q\|_{\text{HS}} \|K\|_2,$$

where we used $\|Q_U\|_{\text{HS}} = \|Q\|_{\text{HS}}$ and $\|K_U\|_2 = \|K\|_2$. Multiplying by $d^{-1/2}$ gives the logit bound. The attention-output bound follows from Proposition 2. $\square$

Proposition 6 (Probabilistic subspace embedding guarantee). Let $U \subseteq \mathcal{H}_d$ be an r -dimensional subspace. Let $S \in \mathbb{R}^{d \times s}$ have i.i.d. entries $S_{ij} \sim \mathcal{N}(0, 1/s)$. For $\epsilon \in (0, 1)$ and $\delta \in (0, 1)$, if

$$s \geq C \frac{r + \ln(1/\delta)}{\epsilon^2},$$

for a universal constant C , then with probability at least $1 - \delta$, S is an ϵ -subspace

embedding for U .

Proof. Let $U_0 \in \mathbb{R}^{d \times r}$ be a matrix whose columns form an orthonormal basis for U . Then S is an ϵ -subspace embedding for U if and only if $\|U_0^T S S^* U_0 - I_r\|_2 \leq \epsilon$. The matrix $U_0^T S S^* U_0 = (S^* U_0)^* (S^* U_0)$ has the same distribution as $G^* G/s$ where $G \in \mathbb{R}^{s \times r}$ has i.i.d. standard normal entries. Standard JL concentration [2] gives $\Pr(\|G^* G/s - I_r\|_2 > \epsilon) \leq 2 \exp(-c s \epsilon^2 + r)$ for a universal constant $c > 0$, provided s is large enough relative to r/ϵ^2 . The stated condition with an appropriate C ensures the failure probability is at most δ . $\square$

Proposition 7 (Finite-precision error for sketched logits). Let $\hat{\mathcal{L}}_S(Q, K)$ denote the logit matrix computed in floating-point arithmetic with unit roundoff u , and let $\gamma_s := su/(1 - su)$ for $su < 1$. Then

$$\|\tilde{\mathcal{L}}_S(Q, K) - \hat{\mathcal{L}}_S(Q, K)\|_{\text{HS}} \leq \frac{\gamma_s \|QS\|_{\text{HS}} \|KS\|_{\text{HS}}}{\sqrt{d}}.$$

Proof. Write the sketched logit product as $d^{-1/2} AB^*$ with $A = QS \in \mathbb{R}^{n \times s}$ and $B = KS \in \mathbb{R}^{n \times s}$. The standard forward-error bound for matrix multiplication [7] gives

$$|\widehat{AB^*} - AB^*| \leq \gamma_s |A| |B^*|,$$

entrywise in absolute value. Taking the Frobenius (Hilbert–Schmidt) norm and applying the Cauchy–Schwarz inequality for matrices,

$$\|\widehat{AB^*} - AB^*\|_{\text{HS}} \leq \gamma_s \| |A| |B^*| \|_{\text{HS}} \leq \gamma_s \|A\|_{\text{HS}} \|B\|_{\text{HS}}.$$

Multiplying by $d^{-1/2}$ yields the bound. $\square$

Remark 8 (Scope of the finite-precision bound). The bound in Proposition 7 models the error in the final sketched logit product $(QS)(KS)^*$, assuming that QS and KS are given exactly. The experiments measure a broader implementation error that also includes storage quantization and low-precision projection effects. $\lrcorner$

Combining the subspace embedding guarantee with the finite-precision error yields the sketching main theorem.

Theorem 9 (Mixed-precision sketched attention). Under the hypotheses of Propositions 6 and 7, with probability at least $1 - \delta$,

$$\|\mathcal{A}(Q, K, V) - \hat{\mathcal{A}}_S(Q, K, V)\|_{\text{HS}} \leq \frac{1}{\sqrt{d}} \left(\epsilon \|Q\|_{\text{HS}} \|K\|_2 + \gamma_s \|QS\|_{\text{HS}} \|KS\|_{\text{HS}} \right) \|V\|_2.$$

The first term is the sketching error; the second is the finite-precision error.

Proof. By Proposition 2 and the triangle inequality,

$$\|\mathcal{A} - \hat{\mathcal{A}}_S\|_{\text{HS}} \leq (\|\mathcal{L} - \tilde{\mathcal{L}}_S\|_{\text{HS}} + \|\tilde{\mathcal{L}}_S - \hat{\mathcal{L}}_S\|_{\text{HS}}) \|V\|_2.$$

Apply Proposition 5 to the first term and Proposition 7 to the second. $\square$

Remark 10 (Design rule from the sketching theorem). Theorem 9 yields a concrete joint design rule. For float16 ($u_{16} \approx 5 \times 10^{-4}$) and typical sketch dimensions $s \leq 64$, we have $s \cdot u_{16} \leq 0.032$, which is smaller than typical sketch tolerances $\epsilon \geq 0.1$. For float32 ($u_{32} \approx 6 \times 10^{-8}$), the condition is essentially always satisfied. For float8 E4M3 ($u_{\text{fp8}} \approx 0.0625$), even $s = 16$ gives $s \cdot u_{\text{fp8}} \approx 1.0$, which is comparable to ϵ —so the finite-precision term is no longer clearly lower order. $\lrcorner$

4 Numerical Experiments

4.1 Exact Attention Under Precision Policies

For each precision policy, we evaluate the exact attention pipeline

$$Q, K, V \rightarrow L = QK^*/\sqrt{d} \rightarrow P = \Sigma(L) \rightarrow A = PV$$

against a float64 reference. The primary diagnostics are the relative logit error

$$E_L = \frac{\|L - \hat{L}\|_{\text{HS}}}{\|L\|_{\text{HS}}},$$

the mean rowwise softmax error

$$E_P = \frac{1}{n} \sum_{i=1}^n \|p_i - \hat{p}_i\|_1,$$

the relative output error

$$E_A = \frac{\|A - \hat{A}\|_{\text{HS}}}{\|A\|_{\text{HS}}},$$

and the softmax amplification ratio

$$\rho_{\text{softmax}} = \frac{\frac{1}{n} \sum_{i=1}^n \|p_i - \hat{p}_i\|_1}{\frac{1}{n} \sum_{i=1}^n \|\ell_i - \hat{\ell}_i\|_{\infty}}.$$

We test the following policy family: **fp32_reference**, **bf16_safe**, **fp16_safe**, low-precision logit storage, low-precision softmax, and low-precision value accumulation. The “safe” policies keep storage low precision but preserve fp32 accumulation and fp32 softmax. This separation is the core experimental point: the question is not whether one global dtype works, but which *locations* are sensitive. The **fp8_store** policy in the sketching baseline (Table 17) should be interpreted as a floating-point quantization-like storage policy rather than affine integer quantization: it reduces representational precision, but it does not introduce an explicit scale and zero-point. In contrast, an int8 attention policy would require choosing calibration ranges for Q, K, V , possibly per tensor or per channel, and would introduce clipping error in addition to floating-point rounding.

Tables 11 and 12 report representative policy-level numbers from Gaussian and transformer-activation sweeps respectively. On Gaussian data (Table 11), all policies remain low-error because the softmax is not in a numerically sensitive regime: $\rho_{\text{softmax}} < 1$ uniformly, so probability errors are generally damped relative to rowwise logit perturbations. The output error E_A remains of the same order as the local logit/value errors. The differences between “safe” and “low” variants are modest in this regime.

Policy	E_L	E_P	E_A	ρ_{softmax}
fp32_reference	8.2e-8	8.5e-8	1.4e-7	0.41
bf16_safe	2.4e-3	1.9e-3	2.8e-3	0.35
fp16_safe	2.9e-4	2.1e-4	3.2e-4	0.32
bf16_low_logit	2.9e-3	2.3e-3	3.4e-3	0.34
fp16_low_logit	3.5e-4	2.8e-4	4.4e-4	0.33
bf16_low_softmax	2.9e-3	2.7e-3	3.7e-3	0.41
fp16_low_softmax	3.5e-4	3.3e-4	4.7e-4	0.39
bf16_low_value	2.4e-3	1.9e-3	3.5e-3	0.35
fp16_low_value	2.9e-4	2.1e-4	4.4e-4	0.32

Table 11 / Exact-attention errors on synthetic Gaussian data ($n \in \{16, 32\}$, $d = 16$, averaged over 2 seeds). “Safe” policies use low-precision storage with fp32 accumulation and softmax.

Policy	E_L	E_P	E_A	ρ_{softmax}
fp32_reference	1.8e-8	3.0e-8	5.0e-8	4.6e4
bf16_safe	6.1e-4	3.0e-8	1.8e-3	1.9
fp16_safe	3.3e-4	3.0e-8	3.5e-4	3.4
bf16_low_logit	3.9e-4	3.0e-8	1.8e-3	2.8
fp16_low_logit	3.9e-4	3.0e-8	3.5e-4	2.8
bf16_low_softmax	3.9e-4	1.2e-5	1.8e-3	1.1e3
fp16_low_softmax	3.9e-4	1.2e-5	3.4e-4	1.1e3
bf16_low_value	6.1e-4	3.0e-8	1.8e-3	1.9
fp16_low_value	3.3e-4	3.0e-8	3.4e-4	3.4

Table 12 / Smoke-test exact-attention errors on a single attention head extracted from a tiny pretrained model (sshleifer/tiny-distilbert-base-cased, layer 0, head 0; $n = 16$, $d = 1$). Because the head dimension is degenerate, this table serves as a consistency check rather than full real-activation evidence. The softmax amplification ratio ρ_{softmax} spikes under low-precision softmax, consistent with Proposition 3 and Remark 4.

The transformer activations (Table 12) reveal a qualitatively different regime. The baseline ρ_{softmax} for **fp32_reference** is $\sim 10^4$ because both numerator and denominator are extremely small, but the rowwise logit-error denominator is several orders of magnitude smaller than the probability-error numerator. This is a diagnostic artifact of dividing two near-zero quantities. The important comparison is between “safe” policies ($\rho \sim 1\text{--}3$) and “low_softmax” policies ($\rho \sim 10^3$). The low-softmax rows should be interpreted relative to the corresponding low-logit rows: both store logits in low precision, but only the low-softmax policy also performs the softmax reduction in low precision. This difference is enough to raise E_P by two orders of magnitude relative to the safe baseline, confirming that low-precision softmax is the clearest failure mode at the probability-matrix level.

The most important empirical message across both tables is that low-precision softmax is the clearest failure mode at the probability-matrix level, while in the tested small regimes, low-precision storage of Q, K, V with fp32 accumulation remains much safer than low-

precision softmax at the probability-matrix level. This is precisely the kind of location-specific statement that a global dtype comparison would miss. The full exact-attention and residual-depth sweeps are committed in `results/precision_policy_sweep.dev.csv`, `results/precision_policy_transformer.dev.csv`, `results/residual_stack_sweep.dev.csv`, and `results/residual_stack_transformer.dev.csv`.

4.2 Attention Entropy Transition and Quantization

A useful way to expose precision sensitivity inside the attention block itself is to introduce an inverse-temperature parameter β and study

$$P_\beta = \text{softmax}(\beta L), \quad L = QK^*/\sqrt{d}.$$

For small β , the attention rows are diffuse and high-entropy. For large β , the rows become concentrated and approach a winner-take-all map. For finite sequence length this is not a thermodynamic phase transition, but it is a sharp finite-size crossover in the geometry of the attention matrix.

We track the normalized entropy

$$\frac{H_\beta}{\log n} = -\frac{1}{n \log n} \sum_{i,j} p_{ij}^{(\beta)} \log p_{ij}^{(\beta)},$$

the mean maximum attention mass

$$M_\beta = \frac{1}{n} \sum_i \max_j p_{ij}^{(\beta)},$$

the probability error E_P , the output error E_A , and the argmax flip rate

$$R_{\text{flip}} = \frac{1}{n} \#\{i : \arg \max_j p_{ij}^{(\beta)} \neq \arg \max_j \hat{p}_{ij}^{(\beta)}\}.$$

This experiment also clarifies the difference between mixed precision and quantization. Mixed precision changes the floating-point format used at different stages of the computation. Quantization introduces an explicit map from real values to a discrete low-bit representation. In the symmetric int8 model used here,

$$q = \text{clip}(\text{round}(x/S), -127, 127), \quad \hat{x} = Sq.$$

Dynamic quantization chooses S from the current tensor range, while the static-logit policy fixes the scale using the $\beta = 1$ logits and reuses it for all β . The latter deliberately exposes clipping once the inverse temperature becomes large. In attention, quantization can be placed on Q, K, V , on the logits, on the probabilities, or on the output; here we test QKV quantization and logit quantization.

Table 13 shows representative rows from the entropy-transition sweep. As β increases, $H/\log n$ decreases from near-unity to near-zero while M_β increases from near-1/ n to near-unity: this is the finite-size crossover from diffuse to concentrated attention. Safe floating-point policies remain stable through this transition: E_P and E_A increase by less than an order of magnitude and R_{flip} stays at zero. Low-precision softmax shows modestly elevated errors but no catastrophic breakdown.

Policy / β	$H/\log n$	M_β	E_P	E_A	R_{flip}	Clip
fp16_safe, $\beta = 0.5$	0.975	0.025	1.12e-4	2.64e-4	0.000	—
fp16_safe, $\beta = 4$	0.335	0.534	6.72e-4	6.80e-4	0.000	—
fp16_safe, $\beta = 16$	0.062	0.883	7.83e-4	1.15e-3	0.000	—
fp16_low_softmax, $\beta = 0.5$	0.975	0.025	2.23e-4	3.65e-4	0.004	—
fp16_low_softmax, $\beta = 4$	0.335	0.534	1.24e-3	1.27e-3	0.004	—
fp16_low_softmax, $\beta = 16$	0.062	0.883	1.55e-3	2.11e-3	0.004	—
int8_logits_static_beta1, $\beta = 0.5$	0.975	0.025	7.56e-3	8.66e-3	0.047	0.000
int8_logits_static_beta1, $\beta = 4$	0.335	0.534	1.44e+0	9.52e-1	0.922	0.322
int8_logits_static_beta1, $\beta = 16$	0.062	0.883	1.92e+0	9.92e-1	0.980	0.804

Table 13 / Representative attention entropy-transition results on Gaussian data ($n = 128$, $d = 32$, averaged over 2 seeds). The full sweep is committed in `results/attention_temperature_sweep.dev.csv`.

The static int8 logit policy tells a different story. At $\beta = 0.5$ the logits lie within the calibration range inherited from $\beta = 1$, clipping is zero, and errors are small. By $\beta = 4$ the rescaled logits βL exceed the calibration range for roughly 32% of entries, E_P exceeds unity, and R_{flip} reaches 92%. At $\beta = 16$ clipping affects 80% of entries and the argmax flip rate is 98%—the quantized attention pattern has almost no overlap with the reference. This is precisely the fragility predicted by the quantization model: once the scale is miscalibrated, clipping introduces large structured errors that the perturbation bound in Proposition 3 does not capture, because $\Delta_{\text{quant}} L$ is no longer small.

The transition experiment thus gives a concrete phase-diagram view of precision placement: low precision is easiest in the diffuse regime and most fragile near concentrated, winner-take-all attention.

4.3 Residual Depth Propagation

To test how local precision errors accumulate across depth, we evaluate a toy attention-only residual dynamical system of the form

$$x_{\ell+1} = x_\ell + hF_\ell(x_\ell)$$

with self-attention blocks F_ℓ . This is a toy stability test—not a full transformer block, since it has no learned projections, MLP, or layer norm—but it isolates how the residual mechanism propagates local precision errors. The metric of interest is the relative terminal-state error

$$E_{\text{depth}} = \frac{\|x_L - \hat{x}_L\|_{\text{HS}}}{\|x_L\|_{\text{HS}}}.$$

Table 14 reports E_{depth} for both Gaussian and transformer-activation residual stacks. On Gaussian data, error grows modestly with depth for all policies: the fp16-safe error roughly doubles from $L = 4$ to $L = 8$, while the bf16-safe error grows by roughly 50%. The low-precision value policies show the steepest depth growth, consistent with the $\sqrt{n}\|\Delta V\|_{\text{HS}}$ term in Proposition 3 accumulating across layers. On transformer activations, the residual scaling $h = 1/L$ keeps errors approximately constant or even decreasing with depth, reflecting the stabilizing effect of small residual contributions.

Policy	Gaussian		Transformer	
	depth 4	depth 8	depth 2	depth 4
fp32_reference	7.2e-8	8.8e-8	5.2e-8	2.2e-8
bf16_safe	3.3e-3	4.7e-3	3.4e-3	1.1e-3
fp16_safe	4.5e-4	6.7e-4	1.6e-4	1.7e-4
bf16_low_logits	3.3e-3	4.9e-3	3.4e-3	1.1e-3
fp16_low_logits	4.5e-4	6.3e-4	1.6e-4	1.7e-4
bf16_low_softmax	3.6e-3	4.7e-3	3.4e-3	1.1e-3
fp16_low_softmax	4.3e-4	6.8e-4	1.6e-4	1.7e-4
bf16_low_value	3.6e-3	5.3e-3	3.1e-3	1.2e-3
fp16_low_value	5.3e-4	7.8e-4	6.6e-4	6.1e-4

Table 14 / Relative terminal-state error E_{depth} for toy attention-only residual dynamical systems (no learned projections, MLP, or layer norm). Gaussian data, $n = d = 16$, averaged over 2 seeds. Depth $L = 4$ uses $h = 0.25$; depth $L = 8$ uses $h = 0.125$. The transformer column uses the same tiny-model smoke-test head as Table 12.

4.4 Ising Magnetization as a Learnability-Transition Test

To study whether reduced precision creates a training-time barrier, we add a toy Ising magnetization task. Each data point is an $L \times L$ spin configuration

$$X \in \{-1, +1\}^{L \times L},$$

sampled from the two-dimensional Ising model at a range of temperatures around the critical value $T_c \approx 2.269$. The normalized magnetization is

$$m(X) = \frac{1}{L^2} \sum_{i,j} X_{ij} \in [-1, 1],$$

and the regression target is the order parameter

$$y = |m(X)| \in [0, 1].$$

Each lattice site is treated as a token containing the spin value and its normalized two-dimensional position. A small one-head attention regressor ($d_{\text{model}} = 32$, depth 1) is trained to predict $|m(X)|$ via MSE loss.

This task is intentionally simple: a full model could compute magnetization by averaging. Its purpose is diagnostic rather than architectural. It gives a controlled setting in which microscopic spins must be aggregated into a macroscopic order parameter, while the data distribution itself changes sharply near the physical critical temperature.

Sanity check: full observation

When all spins are visible, every precision and sketch policy succeeds. Table 15 reports the result: all runs achieve $R^2 \geq 0.97$, and the learned attention remains uniformly diffuse ($H_{\text{attn}}/\log N \approx 1.0$, $M_{\text{attn}} \approx 1/N$). The target can be recovered by global averaging, so no policy fails. Because the unmasked task is solved by high-entropy uniform attention, any subsequent failure on a masked variant can be attributed to the harder inference problem rather than an implementation bug.

Train policy	Eval policy	Test MSE	R^2	Success
fp32	fp32	2.8e-3	0.976	1
fp16_safe	fp16_safe	2.8e-3	0.975	1
int8_qkv_dyn	int8_qkv_dyn	2.9e-3	0.975	1
sketch_4	sketch_4	2.7e-3	0.977	1
sketch_16	sketch_16	3.0e-3	0.974	1
fp32	fp16_safe_eval	3.8e-3	0.967	1
fp32	sketch_4_eval	3.8e-3	0.967	1

Table 15 / Unmasked Ising magnetization. All policies succeed ($R^2 \geq 0.97$). The learned attention is uniformly diffuse, confirming that the task requires only global averaging.

Masked-spin magnetization prediction

To expose a learnability barrier, we randomly mask each spin with probability $1 - p$. The model receives

$$\tilde{X}_{ij} = M_{ij} X_{ij}, \quad M_{ij} \sim \text{Bernoulli}(p),$$

and each token encodes $[\tilde{s}_{ij}, M_{ij}, \text{row}_{ij}, \text{col}_{ij}]$, so the model can distinguish “masked” from “physical spin zero.” The target remains $y = |m(X)|$, computed from the *full* configuration.

We evaluate separately near the critical region $T \in [2.1, 2.5]$, where configurations fluctuate most and the inference problem is hardest. Success requires $R^2_{\text{near } T_c} \geq 0.80$ or $\text{MAE}_{\text{near } T_c} \leq 0.08$.

p	Train	Eval	R^2_{all}	$R^2_{T_c}$	MAE_{T_c}	Succ
0.8	fp32	fp32	0.967	0.935	0.047	1
0.8	fp16_safe	fp16_safe	0.968	0.936	0.046	1
0.8	sketch_4	sketch_4	0.968	0.924	0.051	1
0.4	fp32	fp32	0.941	0.895	0.060	1
0.4	fp16_safe	fp16_safe	0.940	0.895	0.060	1
0.4	sketch_4	sketch_4	0.934	0.882	0.061	1
0.2	fp32	fp32	0.874	0.778	0.085	0
0.2	sketch_4	sketch_4	0.892	0.814	0.080	1
0.2	fp32	sketch_4_eval	0.868	0.733	0.100	0
0.1	fp32	fp32	0.769	0.576	0.111	0
0.1	sketch_4	sketch_4	0.790	0.580	0.116	0
0.1	fp32	sketch_4_eval	0.754	0.501	0.135	0

Table 16 / Masked Ising magnetization prediction ($L = 12$, averaged over 2 seeds). $R^2_{T_c}$ and MAE_{T_c} are evaluated near the critical region. The full sweep is committed in `results/ising_masked_transition.dev.csv`.

Table 16 reveals a finite-size learnability crossover. At $p = 0.8$ and $p = 0.4$, all policies succeed; the task is still easy enough that even aggressive sketching preserves the global aggregation. At $p = 0.2$, a barrier appears: the fp32 model trained from scratch achieves

$R_{T_c}^2 = 0.78$, just below the success threshold, while `sketch_4` trained from scratch reaches $R_{T_c}^2 = 0.81$ and succeeds. Paradoxically, evaluating the trained fp32 model under sketch compression (`sketch_4_eval`) drops $R_{T_c}^2$ to 0.73, worse than training with the sketch from the beginning. At $p = 0.1$, all policies fail near T_c ; the information deficit is too large for any attention policy to overcome.

The $p = 0.2$ result is the clearest evidence of a learnability barrier: the same sketch approximation that harms inference when applied to a pretrained model can *help* training when present from the start, perhaps because the randomized perturbation acts as a regularizer that prevents the optimizer from settling into a brittle, overfitted attention pattern. This is consistent with the wider observation that noise during training can improve generalization in difficult regimes. The barrier is not a thermodynamic phase transition—it is a finite-size crossover in training success—but it has the same qualitative structure: a sharp change in learnability as a control parameter (mask probability) crosses a threshold.

4.5 A Stress-Test Baseline

The experiments in this subsection probe the decomposition of Theorem 9 directly. For each combination of parameters (n, d, s) and each precision configuration we compute three logit matrices:

- (1) $\mathcal{L}(Q, K)$: exact unsketched logits in `float64` (reference);
- (2) $\tilde{\mathcal{L}}_S(Q, K)$: exact sketched logits in `float64`;
- (3) $\hat{\mathcal{L}}_S(Q, K)$: sketched logits in the target precision.

The sketching error is $E_{\text{sk}} := \|\mathcal{L} - \tilde{\mathcal{L}}_S\|_{\text{HS}}$ and the finite-precision error is $E_{\text{fp}} := \|\tilde{\mathcal{L}}_S - \hat{\mathcal{L}}_S\|_{\text{HS}}$. The total error is $E_{\text{tot}} := \|\mathcal{L} - \hat{\mathcal{L}}_S\|_{\text{HS}}$. The same decomposition is recorded for the attention output. We test four precision configurations: `fp32` (all stages in float32), `full_fp16` (storage, projection, final logit product, and softmax in float16), `mixed_fp16` (storage and projection in float16, followed by a float32 final logit product and float32 softmax), and `fp8_store` (storage and projection in float8 E4M3, followed by a float32 final logit product and float32 softmax).

The sketch $S \in \mathbb{R}^{d \times s}$ is a scaled Gaussian matrix with i.i.d. $\mathcal{N}(0, 1/s)$ entries; the scaling is chosen so that $\mathbb{E}[SS^*] = I_d$. We sweep $n \in \{64, 128, 256\}$, $d \in \{32, 64\}$, $s \in \{8, 16, 24, 32\}$, and average over three seeds. Data and sketch seeds are independent. The full numerical results are available in the committed CSV file `results/attention_sweep.csv`.

Since the synthetic Q and K are i.i.d. Gaussian, their joint row space is typically full-dimensional, so for $d = 64$ and $s \leq 32$ the subspace-embedding regime from Proposition 6 is not expected to hold. The experiment should therefore be read as an intentionally under-sketched stress test of the error decomposition, not as evidence that such small feature sketches preserve attention accurately.

Table 17 reports absolute HS errors of the logits for $n = 256$, $d = 64$, together with the ratio $E_{\text{sk}}/E_{\text{fp}}$; the output errors showed the same qualitative ordering and are omitted.

Three observations stand out. First, for fp32 and fp16, $E_{\text{sk}}/E_{\text{fp}}$ is consistently large (10^3 – 10^6), confirming that the dominant error source is the randomized sketch. Second, `full_fp16` and `mixed_fp16` have finite-precision errors of the same order, consistent

Config	s	E_{sk}	E_{fp}	E_{tot}	$E_{\text{sk}}/E_{\text{fp}}$
fp32	8	7.22e2	1.2e-4	7.22e2	5.9e6
	16	5.03e2	1.1e-4	5.03e2	4.4e6
	24	4.27e2	9.4e-5	4.27e2	4.5e6
	32	3.70e2	9.1e-5	3.70e2	4.1e6
full_fp16	8	7.22e2	4.1e-1	7.22e2	1.8e3
	16	5.03e2	2.9e-1	5.03e2	1.8e3
	24	4.27e2	2.6e-1	4.27e2	1.7e3
	32	3.70e2	2.3e-1	3.70e2	1.6e3
mixed_fp16	8	7.22e2	3.8e-1	7.22e2	1.9e3
	16	5.03e2	2.6e-1	5.03e2	1.9e3
	24	4.27e2	2.4e-1	4.27e2	1.8e3
	32	3.70e2	2.1e-1	3.70e2	1.8e3
fp8_store	8	7.22e2	5.0e1	7.21e2	1.4e1
	16	5.03e2	3.4e1	5.01e2	1.5e1
	24	4.27e2	3.0e1	4.26e2	1.4e1
	32	3.70e2	2.6e1	3.70e2	1.4e1

Table 17 / Hilbert–Schmidt errors of the logit matrix ($n = 256$, $d = 64$, averaged over 3 seeds).
 $E_{\text{sk}} = \|\mathcal{L} - \tilde{\mathcal{L}}_S\|_{\text{HS}}$; $E_{\text{fp}} = \|\tilde{\mathcal{L}}_S - \hat{\mathcal{L}}_S\|_{\text{HS}}$; $E_{\text{tot}} = \|\mathcal{L} - \hat{\mathcal{L}}_S\|_{\text{HS}}$; $\|\mathcal{L}\|_{\text{HS}} \approx 256$.

with Proposition 7: storage quantization dominates any benefit from higher-precision accumulation. Third, fp8 storage substantially increases the finite-precision error (from ~ 0.3 to ~ 30 in absolute terms), but the sketching error still dominates by a factor of ~ 14 – 15 . The results support the error decomposition and illustrate the degradation predicted by Theorem 9 for increasing $s \cdot u$, but they do not demonstrate a high-quality efficient-attention approximation or a full inversion of the error ratio.

5 Outlook

The main contribution of this report is the precision-placement decomposition of Proposition 3 and its empirical validation: low-precision softmax is the clearest failure mode in exact attention, while low-precision storage of Q, K, V with fp32 accumulation is typically safe. The sketching baseline (Theorem 9) confirms that the finite-precision term remains lower order as long as $s \cdot u \ll \epsilon$. Mixed precision and quantization are both precision-placement mechanisms, but quantization adds calibration and clipping error; the same local perturbation framework applies to both, provided the quantized tensor is modeled as $X + \Delta X$, where ΔX is controlled not only by unit roundoff but also by scale, zero-point, range selection, and saturation.

Several directions remain open. First, the current experiments use small problem sizes ($n \leq 256$, $d \leq 64$); scaling to realistic transformer dimensions will require either GPU acceleration or the JAX backend scaffold already present in the repository. Second, FP8 policies are reported only under an explicit quantization model; a more complete treatment would include hardware-specific scaling and dequantization. Third, the phase-diagram analogy from Decelle et al. [8] suggests mapping not only average error but also regime changes: when does low-precision logit storage remain harmless, when does

low-precision softmax begin to amplify, and when does no practical low-precision policy preserve the reference behavior? The current data already hints at these transitions but does not yet chart them systematically. Fourth, the JAX-focused mixed-precision framework of Graefe and Trimpe [9] provides a natural implementation path for the safe/unsafe policy distinctions identified here. We note that the current JAX backend in the repository maps PyTorch fp8 dtypes to JAX float16, so JAX-based fp8 experiments would require a dedicated quantization path and should not be compared directly with PyTorch fp8 results. Fifth, the masked Ising experiment reveals a finite-size learnability crossover at $p \approx 0.2$: below this mask probability, no policy can reliably predict the magnetization near T_c , while above it, all policies succeed. The sketch-trained model at $p = 0.2$ succeeds where the fp32-trained model evaluated under sketch compression fails, suggesting a training-time regularization effect. A more systematic sweep of mask probabilities and model depths would clarify whether this crossover sharpens into a true threshold.

On the analytical side, the $\sqrt{n}$ bound on $\|\Sigma(\hat{L})\|_2$ is likely loose for near-doubly-stochastic attention matrices; a refined bound that exploits the concentration of softmax rows could tighten the value-perturbation term. The logit-perturbation term could similarly be refined by replacing the worst-case $\|V\|_2$ with an average-case or data-dependent quantity.

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